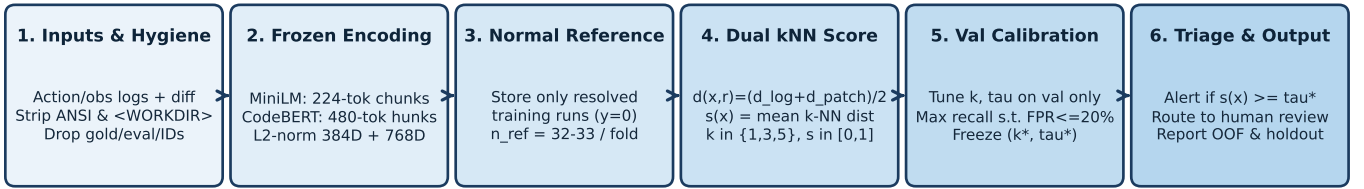
1. Inputs & Hygiene
Action/obs logs + diff
Strip ANSI & <WORKDIR>
Drop gold/eval/IDs
2. Frozen Encoding
MiniLM: 224-tok chunks
CodeBERT: 480-tok hunks
L2-norm 384D + 768D
3. Normal Reference
Store only resolved
training runs (y=0)
n_ref = 32-33 / fold
4. Dual kNN Score
d(x,r)=(d_log+d_patch)/2
s(x) = mean k-NN dist
k in {1,3,5}, s in [0,1]
5. Val Calibration
Tune k, tau on val only
Max recall s.t. FPR<=20%
Freeze (k*, tau*)
6. Triage & Output
Alert if s(x) >= tau*
Route to human review
Report OOF & holdout